

LawPal

A Retrieval Augmented Generation Based System for Enhanced Legal
Accessibility in India

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Problem :

Legal information in India is often complex, fragmented, and difficult for the average person to navigate, leading to misunderstandings and barriers to justice.

Solution :

develop a RAG-based AI system that provides context-aware, accurate, and easily understandable responses to legal queries, specifically tailored for Indian laws and regulations.

RAG pipeline

A Retrieval-Augmented Generation (RAG) pipeline enhances the capabilities of large language models (LLMs) by giving them access to external, up-to-date, and domain-specific information, thereby improving the accuracy and relevance of their responses. It works by combining an information retrieval system with a text generator.

How a RAG Pipeline Works

The RAG pipeline typically involves two main phases: retrieval and generation.

1. Retrieval Phase

When a user asks a question, the RAG system does not immediately send the prompt to the language model. Instead, it first searches a vast, curated knowledge base (such as a database of documents, articles, or internal company data) to find relevant context

The steps involved are:

- ***Indexing and Embedding:*** *The external knowledge base is pre-processed and indexed. Each document or chunk of text is converted into a numerical representation called a vector embedding, which captures its semantic meaning.*
- ***Vector Search:*** *The user's query is also converted into a vector embedding. A vector similarity search algorithm is then used to quickly find the top "k" most relevant documents in the knowledge base that are semantically similar to the query.*

2. Generation Phase

The retrieved relevant documents are then combined with the original user query to form a new, more informed prompt. This enriched prompt is sent to the large language model (like GPT-4, Llama, etc.)

The steps involved are:

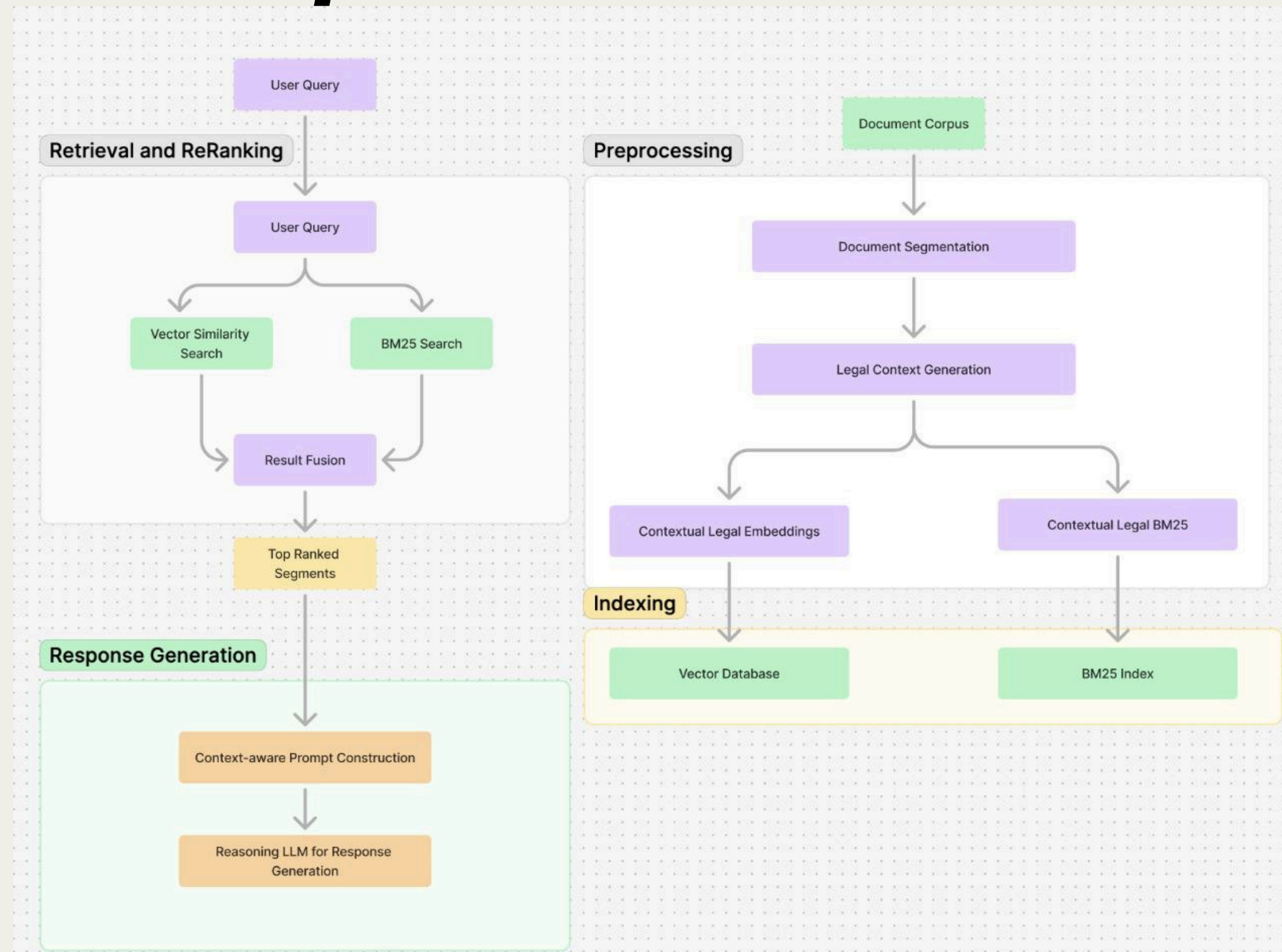
- **Prompt Engineering:** *The system creates a comprehensive prompt that essentially instructs the LLM: "Using the following context, please answer the user's question..."*
- **Augmented Generation:** *The LLM uses the provided context to generate an accurate and relevant answer. Because the model is grounded in the retrieved facts, it is less likely to "hallucinate" (generate false information) and can provide answers based on specific, verifiable data.*

Benefits of RAG

The primary advantages of using a RAG pipeline include:

- **Improved Accuracy:** *RAG reduces the likelihood of the LLM generating incorrect or irrelevant information by providing factual grounding*
- **Access to Current Data:** *It allows the LLM to access and use information that was not included in its initial training data*
- **Domain Specificity:** *RAG enables the model to answer questions about proprietary or specialized information (e.g., internal company policies, specific research papers)*
- **Transparency:** *RAG systems can often cite their sources, allowing users to verify the information provided*

RAG Pipeline Architecture



Workflow of the Project

- ***Daily Scraping of Laws***
- ***Cleaning & Chunking***
- ***Embedding & Index Creation***
- ***Query → Retrieval → Generation***
- ***Display Response to User***

Communication flow of the Project

- ***Frontend***
- ***FastAPI***
- ***Retrieval (ChromaDB + BM25)***
- ***LLM***
- ***Response***
- ***Frontend***

Underlying data and Processing

Dataset Sources:

The system gathers legal information from authoritative sources to ensure accuracy and relevance.

- *Government law portals (e.g., India Code, Legal Information portals)*
- *Periodically scraped updates via the automated crawler*

Processing Pipeline:

A systematic pipeline transforms raw data into a structured and searchable format for the RAG system.

- ***Raw HTML → Text extraction (BeautifulSoup)***
- ***Cleaning → Lowercasing, token removal***
- ***Chunking (split long sections into smaller units)***
- ***Embedding (sentence transformer model)***
- ***Storage in ChromaDB + BM25 index***

Mathematical & Algorithmic Foundation

BM25 Keyword Retrieval:

Ranks documents by relevance based on term frequency and inverse document frequency.

$$\text{Score}(D, Q) = \frac{\sum_{i=1}^n \text{IDF}(q_i) \cdot f(q_i, D)(k_1 + 1)}{f(q_i, D) + k_1(1 - b + b \cdot |D|/\text{avgD})}$$

Vector Similarity Search (ChromaDB):

Uses cosine similarity between embedding vectors of the query and document chunks.

$$\text{similarity}(A, B) = \frac{A \cdot B}{\|A\| \|B\|}$$

Searching Mechanism

Brain 1: The Keyword Search (BM25)

What it does: It's like a classic "Ctrl+F." It's fast and perfect for finding exact words, like "Section 181" or "Indian Evidence Act."

Brain 2: The Meaning Search (Vector Search)

What it does: This search understands concepts. It knows that a user's scenario like "my cheque bounced" has the same meaning as the legal text "dishonour of cheque."

*How it works: We use the powerful *BAAI/bge-large-en-v1.5* model (our "translator") to turn all chunks into numbers (vectors). We store these in a ChromaDB database. When you ask a question, we translate the query into numbers and find the closest match.*

The Problem: Our Hybrid Search gives us a list of 20+ possible answers. But the best answer might be buried at #15, and a wrong answer might be at #1.

Our Solution: We built an Advanced Legal Reranker. This is a smart filter that re-sorts the 20+ results based on a 4-part scoring system to find the perfect chunk and move it to the top.

Our 4-Part Scoring System:

- **Entity Match:** If you ask for "Section 32 of the Indian Evidence Act," this rule gives a *massive 10x bonus* to that exact chunk. This ensures specific questions are answered with perfect precision.
- **Semantic Score:** How closely does the meaning of the chunk match the meaning of your query? (This is the main score).
- **Keyword Boost:** A small bonus for chunks that contain important keywords from your query, like "penalty," "insurance," or "fine." This helped us fix the "uninsured vehicle" bug.
- **Heuristic Bonus:** A tiny "tie-breaker" bonus for foundational laws like the Constitution or Indian Penal Code.

This reranker is what makes our system so accurate.

Web Scraper

A web scraper is an automated tool that extracts data from websites, saving it in a structured format like a spreadsheet or database for later analysis

How it works

- *Crawling: A web crawler navigates the web by following links and fetching pages to be scraped.*
- *Parsing: The scraper then analyzes the page's HTML code to find and extract specific data points.*
- *Extraction: The extracted information, such as names, prices, or addresses, is copied into a usable format.*
- *Export: Finally, the data is exported to a local file, like a spreadsheet or a database.*

Tech Stack

Frontend Technology Stack:

React 18 with TypeScript

Tailwind CSS for styling

Vite for dev server & build

Framer Motion for animations

React Markdown for chat formatting

JWT for authentication

Backend Technology Stack:

Node.js + Express

TypeScript

MongoDB Atlas (cloud database)

JWT authentication

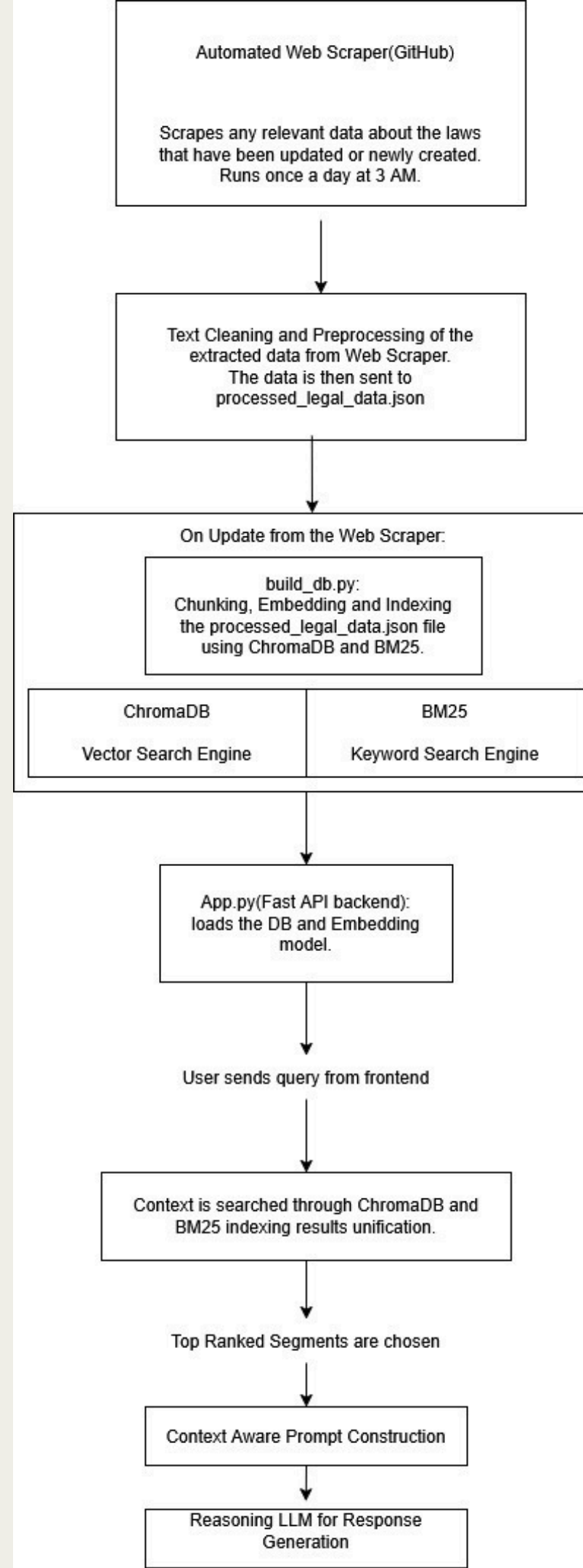
SendGrid email service (HTTPS API)

Groq API integration

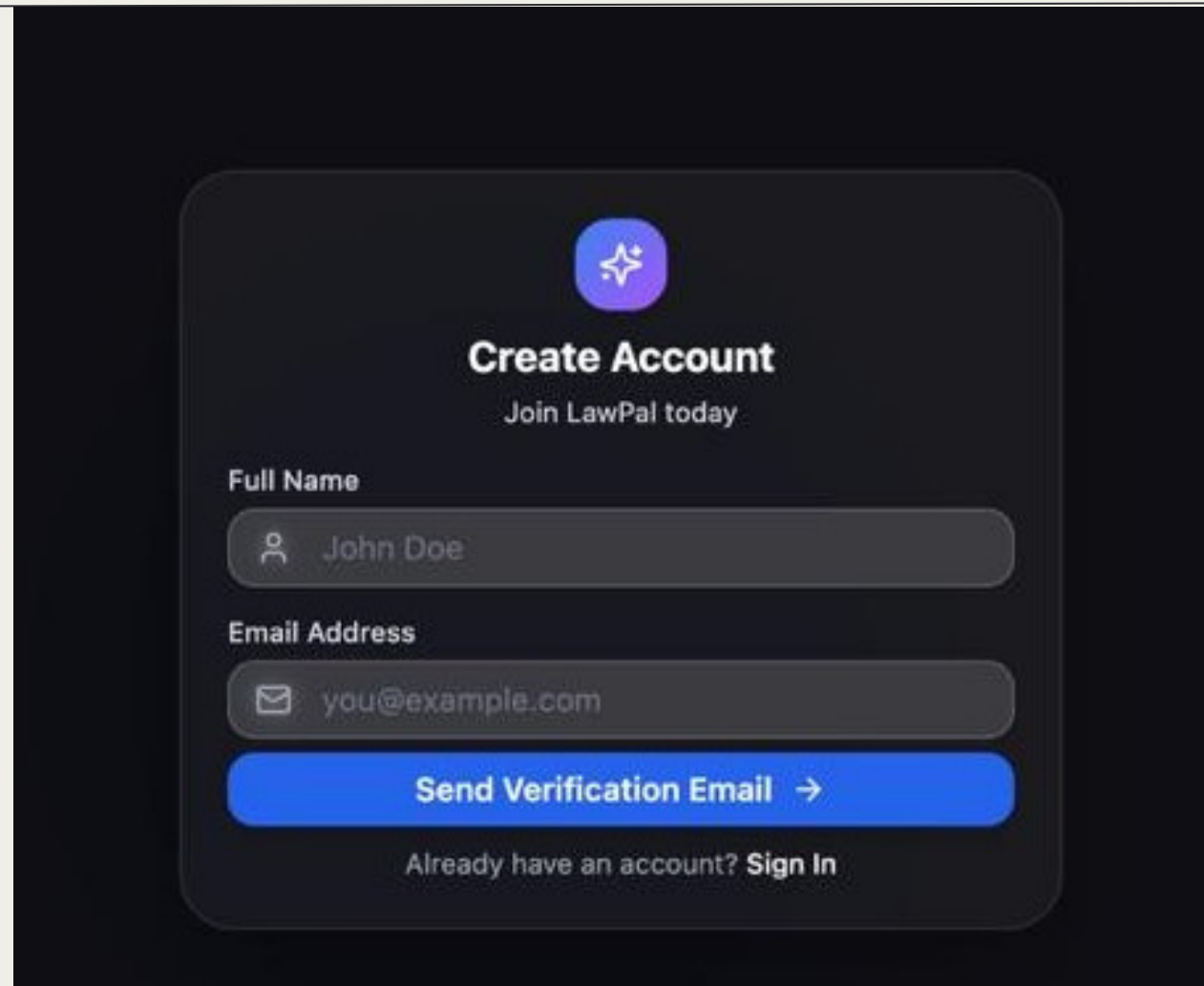
bcrypt for password hashing

CORS configured for multiple origins

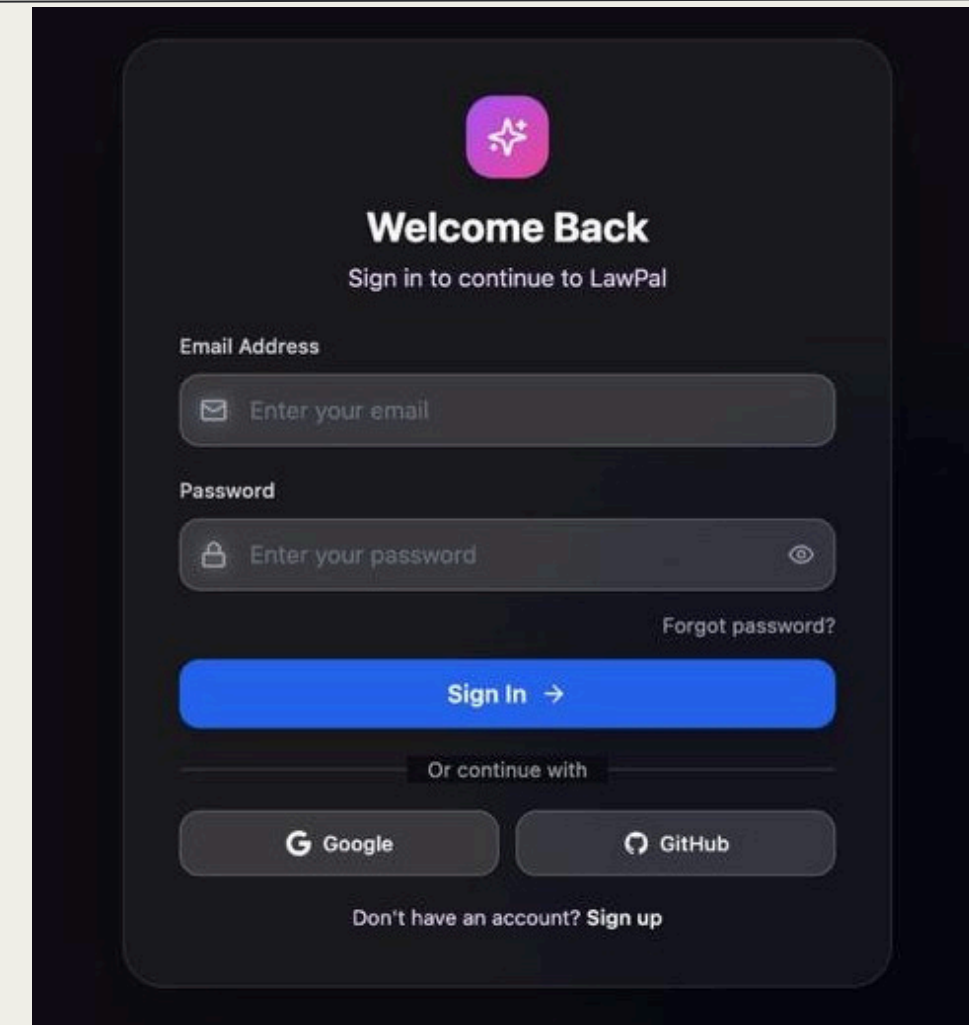
Architecture Diagram



User Authentication



The 'Create Account' form features a purple star icon at the top. Below the title, it prompts the user to 'Join LawPal today'. The form includes two input fields: 'Full Name' with the placeholder 'John Doe' and 'Email Address' with the placeholder 'you@example.com'. A prominent blue button labeled 'Send Verification Email →' is positioned below the email field. At the bottom, a link states 'Already have an account? Sign In'.



The 'Welcome Back' form features a purple star icon at the top. Below the title, it prompts the user to 'Sign in to continue to LawPal'. The form includes two input fields: 'Email Address' with the placeholder 'Enter your email' and 'Password' with the placeholder 'Enter your password' and a toggle icon. A link for 'Forgot password?' is located to the right of the password field. A prominent blue button labeled 'Sign In →' is positioned below the password field. Below this, a section titled 'Or continue with' contains two buttons for 'Google' and 'GitHub'. At the bottom, a link states 'Don't have an account? Sign up'.

LawPal provides secure user access through flexible login options, including Google OAuth, login using github as well as traditional email/password, all managed by NextAuth. This ensures only authorized users can access the AI-powered legal analysis tools.

A consistent user ID is maintained regardless of the login method chosen. This allows all generated summaries, and Q&A history to be securely associated with the user, enabling a personalized and persistent experience across sessions.



Create Account

Join LawPal today

Full Name



John Doe

Email Address



you@example.com

Send Verification Email →

Already have an account? **Sign In**



Welcome Back

Sign in to continue to LawPal

Email Address



Enter your email

Password



Enter your password



[Forgot password?](#)

Sign In →

Or continue with



Google



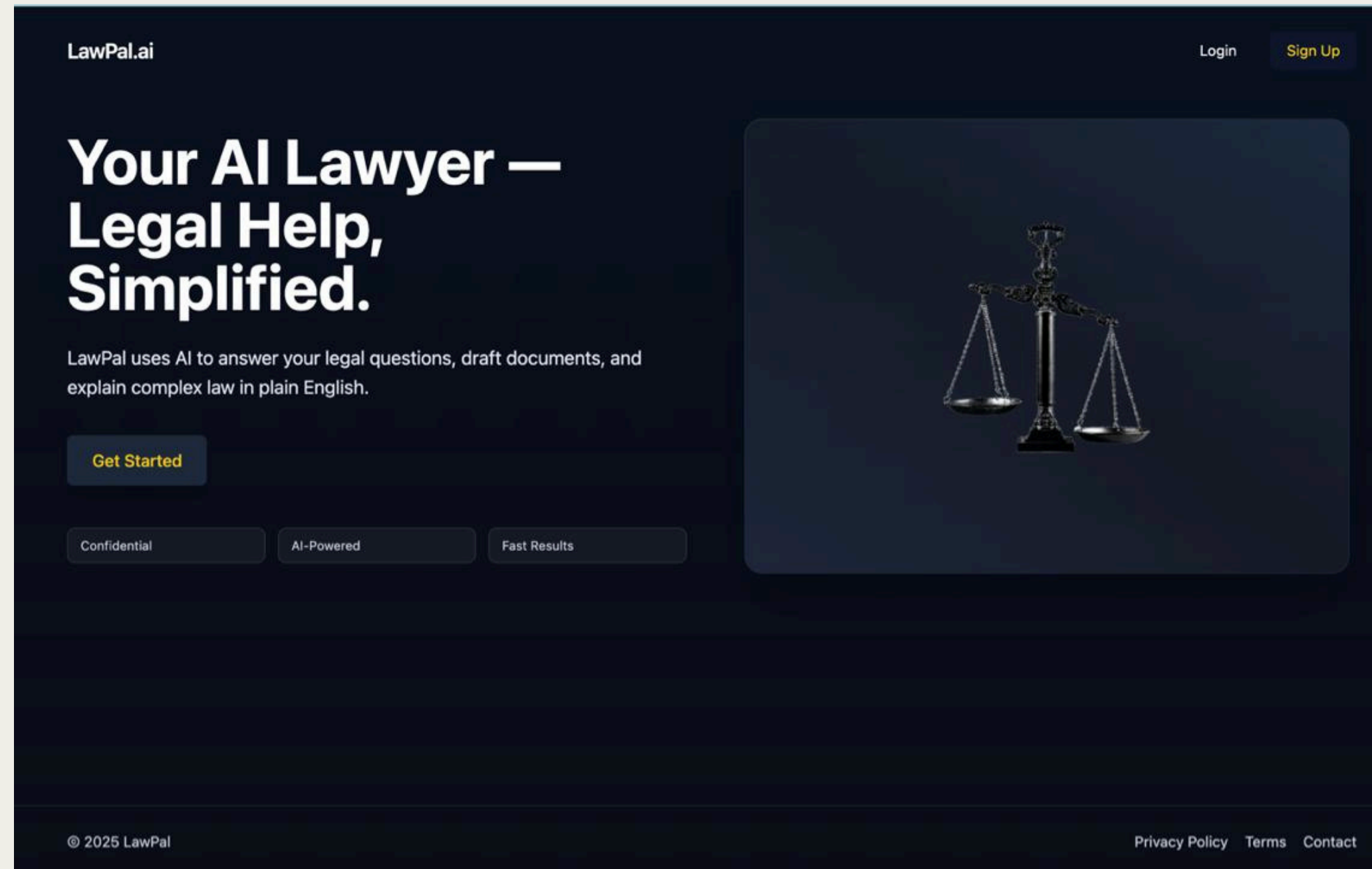
GitHub

Don't have an account? **Sign up**

Frontend

Our user-friendly frontend, built with Next.js and React, serves as the primary interface for LawPal. It's designed for seamless interaction with our AI-powered legal analysis tools.

All these capabilities – document management, summary review, and question answering – are cohesively integrated. This provides centralized access within an authenticated and personalized environment, streamlining the user's interaction with complex legal information.





AI Lawyer

Your Personal Legal AI



+ New Conversation

Sai stabbed a man with a knife; ...

6 messages

11/13/2025, 10:27:47 PM

Explain Section 279 IPC

2 messages

11/13/2025, 10:25:31 PM

If someone causes death by acci...

14 messages

11/13/2025, 8:59:54 PM

If someone causes death by acci...

4 messages

11/13/2025, 8:56:08 PM

What is Section 302 in IPC?

4 messages

11/13/2025, 8:53:54 PM

how to save my friend

2 messages

11/13/2025, 8:52:44 PM

hi

8 messages

11/13/2025, 8:51:34 PM

hello

10 messages

11/12/2025, 11:06:35 PM

hello

24 messages

11/12/2025, 9:26:23 PM



Sainath

abcd1@gmail.com

Sai stabbed a man with a knife; what punishment could he ...

6 messages



under this section. However, if the stab wound caused grievous hurt, Sai could face punishment under Section 326 of the Indian Penal Code, 1860, which prescribes imprisonment for life, or imprisonment of either description for a term which may extend to ten years, and shall also be liable to fine.

Confidence: 8/10 — based on the completeness of the provided legal sources.



Under Section 86 of the Indian Penal Code, 1860, which is not provided in the given sources, it's generally stated that if a person causes hurt or death in a sudden fight without premeditation, the punishment can be less severe. However, the exact details of this section are not mentioned in the provided sources.

But considering the similar context provided in Section 335 of the Indian Penal Code, 1860, which deals with "Voluntarily causing grievous hurt on provocation," if the fight resulted in grievous hurt and was on grave and sudden provocation, the person might be punished with imprisonment for up to four years, or with a fine, or with both.

Additionally, Section 334 of the Indian Penal Code, 1860, mentions that if hurt (not grievous) is caused in a sudden fight on provocation, the punishment could be imprisonment for up to one month, or a fine, or both.

To give a precise answer, more details about the situation, such as the extent of the injury and the nature of the provocation, are needed. Based on the provided sources, it

Can we save him? It was a sudden fight, no planning."



Enter a message, or type : for commands...

Send

Problems we faced

- **Memory Constraints** (Colab): Limited batch/sequence sizes, addressed by optimizing training.
- **Handling long documents:** Necessitated document chunking and pre-summarization techniques.
- **Legal accuracy:** Ensuring legal accuracy in generated outputs.
- **Training Reliability:** Implemented epoch-wise checkpointing to save progress & avoid re-runs.

Use Cases

LawPal offers significant benefits to legal professionals, paralegals, students, and academics by accelerating case assessment, research, and understanding of complex judgments. Lawyers can rapidly grasp case overviews and efficiently locate specific facts or precedents through Q&A, aiding in due diligence and brief preparation. For students and researchers, the system provides concise summaries of dense case law, streamlining the learning and information gathering process.

Future scope

- Future development will focus on enhancing model accuracy and improving our web scraper,
- Training on more specialised legal datasets.
- API integration
- Multilingual support, and user experience improvements
- Interactive visualizations
- Adding voice based query system, aiming to make it an even more powerful and versatile tool for comprehensive legal document analysis.

Project Goals and Conclusions

Initial Goals (Milestone 1):

Document Corpus: Extract & prepare at legal documents.

Core Functionality: Enable users to ask interactive Q&A and successfully build the RAG pipeline.

Model Prototypes: Train working question-answering model.

Final Outcomes Delivered:

Working Prototype: Achieved. Has accurate retrieval over legal data. RAG-based responses are aligned with the legal context.

Core Functionality: Successfully Met. Users can effectively get answers via Q&A.

Model Development: Successfully Trained. Our RAG architecture works exactly as it is designed for.

Thank you!

Demo
